



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Integrated Bachelor's – Master's program

End-term Examination - Autumn 2025

Subject: Introduction to Computing

Subject Code(s): COM 1101

Full marks: 50

Time allotted: 3 hrs

Instruction: Answer Q1 and any four questions from the rest.

Best of luck for your exam!

-
1. Explain the output of the programs. If there is any mistake in any program that may lead to compilation error, mention the necessary correction(s) that will lead to error-free compilation. Attempt any six out of seven. {6 × 3 = 18 Marks}

- i) Refer to the program on pointer and pointer arithmetic -

```
#include <stdio.h>
int main(void)
{
    int a[] = {3, 6, 9, 12};
    int *p = a;
    const int *q = p + 3;
    printf("%d ", *p++);
    printf("%d ", *(&p));
    printf("%d ", *--q);
    printf("%d\n", *p + *q);
    return 0;
}
```

- ii) Refer to the program on dynamic memory allocation and deallocation -

```
#include <stdio.h>
#include <stdlib.h>
int main(void)
{
    int *p = malloc(3 * sizeof(int));
```

```

    for(int i = 0; i < 3; i++) p[i] = i + 1;
    int *q = p;
    p = realloc(p, 5 * sizeof(int));
    for(int i = 3; i < 5; i++) p[i] = p[i-1] + q[i-3];
    printf("%d %d %d %d %d\n", p[0], p[1], p[2], p[3], p[4]);
    return 0;
}

```

iii) Refer to the program on strings in C -

```

#include <stdio.h>
int main(void)
{
    char *p = "hello";
    char q[] = "hello";
    printf("%c ", *p);
    printf("%c ", ++*q);
    p = q + 2;
    q[1] = 'A';
    printf("%s %s\n", p, q);
    return 0;
}

```

iv) Refer to the program on scope and longevity of static variables in C -

```

#include <stdio.h>
int fun(int x)
{
    static int a = 2;
    int b = 1;
    a += x;
    b += x;
    return a + b;
}
int main(void)
{
    int a = 5;
    for(int i = 0; i < 3; i++)
    {
        int a = i;
    }
}

```

```

        printf("%d ", fun(a));
    }
    printf("\n");
    return 0;
}

```

v) Refer to the program on recursion and global variables in C -

```

#include <stdio.h>
int g = 1;
int f(int x)
{
    if (x == 0) return g;
    g += x;
    return f(x - 1) + x;
}
int main(void)
{
    printf("%d\n", f(4));
    return 0;
}

```

vi) Refer to the program on file access in C -

```

#include <stdio.h>
int main(void)
{
    FILE *fp = fopen("input.txt", "r");
    char buf[4];
    int count = 0;
    while (fgets(buf, sizeof(buf), fp))
    {
        count++;
        printf("%s", buf);
    }
    fclose(fp);
    printf("\nLines read in chunks: %d\n", count);
    return 0;
}

```

Say, the file input.txt contains -

ABCDE

XY

123456

vii) Refer to the program on pointer manipulation in C -

```
#include <stdio.h>
#include <stdlib.h>
int rec(int *p, int n)
{
    if (n == 1) return *p;
    return *p + rec(p + 1, n - 1) - rec(p, n - 1);
}
int main(void)
{
    int n = 5;
    int *a = malloc(n * sizeof(int));
    for(int i = 0; i < n; i++) a[i] = i + 1;
    printf("%d\n", rec(a, n));
    free(a);
    return 0;
}
```

2. (a) [3 marks] Explain the following pointer behaviour:

```
int x = 10;
int *p = &x;
int **q = &p;
*p = *p + 5;
**q = **q + 2;
printf("%d\n", x);
```

Explain why the value of x is updated twice even though different pointer expressions are used.

(b) [5 marks] Write a C program that reads and stores N integers using dynamic memory allocation and computes the sum of only the even numbers using a function that receives a pointer to the first element.

3. (a) [3 marks] Explain why `sizeof(arr)` gives different results when `arr` is declared as:

```
int arr[20];
```

versus when it is passed to a function:

```
void func(int arr[])
{
    printf("%lu\n", sizeof(arr));
}
```

- (b) [5 marks] Write a C program to read an array of integers and determine the second largest **distinct** number without using sorting.

4. (a) [3 marks] Explain the following structure access behaviour:

```
struct Point
{
    int x;
    int y;
};
struct Point p = {10, 20};
struct Point *ptr = &p;
ptr->x = ptr->x + 5;
```

Explain why updating `ptr->x` changes `p.x`.

- (b) [5 marks] Write a C program defining a `Student` structure with members `name`, `roll`, and `marks`. Read details of N students and print the record of the student who scored the highest marks.

5. (a) [3 marks] Explain what happens in memory during the following operations:

```
int *p = malloc(sizeof(int));
*p = 50;
p = malloc(sizeof(int));
```

- (b) [5 marks] Write a C program that reads N integers using `malloc`, creates a new dynamically allocated array containing only positive integers, and prints the resulting array.

6. (a) [3 marks] Given the following:

```
char s1[20] = "abc";  
char s2[20] = "abcd";  
printf("%d", strcmp(s1, s2));
```

Explain why the output is negative.

(b) [5 marks] Write a C program that reads a sentence, uses `strtok` to break it into words, and prints each word along with its length (using `strlen`).

7. (a) [3 marks] Explain why the following write operation fails:

```
FILE *fp;  
fp = fopen("data.txt", "r");  
fprintf(fp, "Hello");
```

(b) [5 marks] Write a C program that reads integers from `input.txt`, counts how many of them are prime, and writes the count to `output.txt`.

8. (a) [3 marks] Identify the problem in the following code:

```
char *p = "hello";  
p[0] = 'H';
```

(b) [5 marks] Write a C program that dynamically allocates memory for a string, reads a line of text, and converts all lowercase letters to uppercase using pointer arithmetic (not array indexing).